

Experiment 'p_1_10_animal_speaking' Results

June 3, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_1_10_animal_speaking

Natural language statement: Write a method that returns an animal speaking a greeting in ASCII art.

Method signature: p_1_10_animal_speaking() returns (s: string)

Ensures

- `s == " /_/_/\ /---\\\n (' ') / Hello \\\n (-) | New |\n | | | \\\n Coder /\n (__|__)
\\-----/\n"`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method that returns an animal speaking a greeting in ASCII art.

The signature should be:

`p_1_10_animal_speaking()` returns (`s: string`)

The method should respect the following contract:

ensures `s == " /\_\_/\_\ /---\\\n ( ' ' ) / Hello \\\n ( - ) | New |\n | | | \_\_ Coder /\n (`
↪ `__|__ ) \\\n----/\n"`

Produce and show only the Dafny body of this method, including the curly braces that
↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor
↪ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DmdNbV9O9azJTSHnildpeBVMYasEY

Model: gpt-5.5-2026-04-23

Created at: 1780483491

Finish reason: stop

Usage: 352 tokens in, and 495 tokens out

```
{
  s := " /\_\_/\_\ /---\\\n ( ' ' ) / Hello \\\n ( - ) | New |\n | | | \_\_ Coder /\n (__|__)
  ↪ \\\n----/\n";
}
```

Final Program

```
method p_1_10_animal_speaking() returns (s: string)
  ensures s == " /\_\_/\_\ /---\\\n ( ' ' ) / Hello \\\n ( - ) | New |\n | | | \_\_
  ↪ Coder /\n (__|__) \\\n----/\n"
{
  s := " /\_\_/\_\ /---\\\n ( ' ' ) / Hello \\\n ( - ) | New |\n | | | \_\_ Coder /\n (__|__)
  ↪ \\\n----/\n";
}
```

Total Token Usage

Input tokens: 352

Output tokens: 495

Reasoning tokens: 409

Sum of ‘total tokens’: 847

Experiment Timings

Overall Experiment started at 1780483491263, ended at 1780483508319, lasting 17056ms (17.06 seconds)

Iteration #1 started at 1780483491264, ended at 1780483508318, lasting 17054ms (17.05 seconds)

Experiment 'p_1_1_greeting' Results

June 3, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_1_1_greeting

Natural language statement: Write a method that returns the greeting "Hugo".

Method signature: p_1_1_greeting() returns (s: string)

Ensures

- `s == "Hugo"`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method that returns the greeting "Hugo".

The signature should be:

`p_1_1_greeting()` returns (`s: string`)

The method should respect the following contract:

`ensures s == "Hugo"`

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DmdKyvQkAbTy2rhgCZN6HBnjoUUig

Model: gpt-5.5-2026-04-23

Created at: 1780483328

Finish reason: stop

Usage: 242 tokens in, and 23 tokens out

```
{  
  s := "Hugo";  
}
```

Final Program

```
method p_1_1_greeting() returns (s: string)  
  ensures s == "Hugo"  
{  
  s := "Hugo";  
}
```

Total Token Usage

Input tokens: 242

Output tokens: 23

Reasoning tokens: 0

Sum of ‘total tokens’: 265

Experiment Timings

Overall Experiment started at 1780483325813, ended at 1780483335665, lasting 9852ms (9.85 seconds)

Iteration #1 started at 1780483325821, ended at 1780483335664, lasting 9843ms (9.84 seconds)

Experiment 'p_1_2_sum' Results

June 3, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_1_2_sum

Natural language statement: Write a method that returns the sum of the first ten positive integers, $1 + 2 + \dots + 10$.

Method signature: p_1_2_sum() returns (i: int)

Ensures

- $i == (10 * 11) / 2$

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method that returns the sum of the first ten positive integers, $1 + 2 + \dots + 10$.

The signature should be:

`p_1_2_sum()` returns `(i: int)`

The method should respect the following contract:

`ensures i == (10 * 11) / 2`

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DmdL7pkik34BX8oatarD62p5cbxQ2

Model: gpt-5.5-2026-04-23

Created at: 1780483337

Finish reason: stop

Usage: 280 tokens in, and 85 tokens out

```
{  
  i := (10 * 11) / 2;  
}
```

Final Program

```
method p_1_2_sum() returns (i: int)  
  ensures i == (10 * 11) / 2  
{  
  i := (10 * 11) / 2;  
}
```

Total Token Usage

Input tokens: 280

Output tokens: 85

Reasoning tokens: 51

Sum of ‘total tokens’: 365

Experiment Timings

Overall Experiment started at 1780483337431, ended at 1780483344423, lasting 6992ms (6.99 seconds)

Iteration #1 started at 1780483337432, ended at 1780483344423, lasting 6991ms (6.99 seconds)

Experiment ‘p_1_3_product’ Results

June 3, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_1_3_product

Natural language statement: Write a method that returns the product of the first ten positive integers, $1 * 2 * \dots * 10$.

Method signature: p_1_3_product() returns (i: int)

Ensures

- $i == \text{factorial}(10)$

Functional Code Given

```
function factorial(n: int): int
  decreases n
{
  if n <= 1 then 1 else n * factorial(n - 1)
}
```

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method that returns the product of the first ten positive integers, $1 * 2 * \dots *$
→ 10.

The signature should be:

p_1_3_product() returns (i: int)

The method should respect the following contract:

ensures i == factorial(10)

The contract uses the following dafny code:

```
function factorial(n: int): int
  decreases n
{
  if n <= 1 then 1 else n * factorial(n - 1)
}
```

These function(s) must not be used in your implementation of the method.

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DmdLFPzUk54JeDgiaVWgbHMfcpnSQ

Model: gpt-5.5-2026-04-23

Created at: 1780483345

Finish reason: stop

Usage: 380 tokens in, and 541 tokens out

```
{
  i := 3628800;
}
```

Final Program

```
function factorial(n: int): int
  decreases n
```

```
{  
  if n <= 1 then 1 else n * factorial(n - 1)  
}  
  
method p_1_3_product() returns (i: int)  
  ensures i == factorial(10)  
{  
  i := 3628800;  
}
```

Total Token Usage

Input tokens: 380

Output tokens: 541

Reasoning tokens: 512

Sum of 'total tokens': 921

Experiment Timings

Overall Experiment started at 1780483345230, ended at 1780483364351, lasting 19121ms (19.12 seconds)

Iteration #1 started at 1780483345231, ended at 1780483364350, lasting 19119ms (19.12 seconds)

Experiment 'p_1_4_account_balance' Results

June 3, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_1_4_account_balance

Natural language statement: Write a method that returns the balance of an account after the first, second, and third year. The account has an initial balance of \$1,000 and earns 5 percent interest per year.

Method signature: p_1_4_account_balance() returns (first: real, second: real, third: real)

Ensures

- `first == 1000.0 + 1000.0 * 0.05`
- `second == first + first * 0.05`
- `third == second + second * 0.05`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method that returns the balance of an account after the first, second, and third
→ year. The account has an initial balance of \$1,000 and earns 5 percent interest
→ per year.

The signature should be:

`p_1_4_account_balance()` returns (first: real, second: real, third: real)

The method should respect the following contract:

ensures first == 1000.0 + 1000.0 * 0.05, ensures second == first + first * 0.05, ensures
→ third == second + second * 0.05

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DmdLZLaZ4Zq71WVGy1vZQ7Yu1LI5g

Model: gpt-5.5-2026-04-23

Created at: 1780483365

Finish reason: stop

Usage: 390 tokens in, and 184 tokens out

```
{
  first := 1000.0 + 1000.0 * 0.05;
  second := first + first * 0.05;
  third := second + second * 0.05;
}
```

Final Program

```
method p_1_4_account_balance() returns (first: real, second: real, third: real)
  ensures first == 1000.0 + 1000.0 * 0.05
  ensures second == first + first * 0.05
  ensures third == second + second * 0.05
{
  first := 1000.0 + 1000.0 * 0.05;
  second := first + first * 0.05;
  third := second + second * 0.05;
}
```

}

Total Token Usage

Input tokens: 390

Output tokens: 184

Reasoning tokens: 117

Sum of 'total tokens': 574

Experiment Timings

Overall Experiment started at 1780483365161, ended at 1780483373201, lasting 8040ms (8.04 seconds)

Iteration #1 started at 1780483365163, ended at 1780483373201, lasting 8038ms (8.04 seconds)

Experiment ‘p_1_5_box’ Results

June 3, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_1_5_box

Natural language statement: Write a method that returns the name Dave inside a box on the screen.

Do your best to approximate lines with characters such as | - +.

Method signature: p_1_5_box() returns (s: string)

Ensures

- `s == "+-----+\n| Dave |\n+-----+\n"`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method that returns the name Dave inside a box on the screen. Do your best to
↪ approximate lines with characters such as | - +.

The signature should be:

p_1_5_box() returns (s: string)

The method should respect the following contract:

ensures s == "+-----+\n| Dave |\n+-----+\n"

Produce and show only the Dafny body of this method, including the curly braces that
↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor
↪ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DmdLierRvmNjC2ANt64BKOze7QRUA

Model: gpt-5.5-2026-04-23

Created at: 1780483374

Finish reason: stop

Usage: 294 tokens in, and 97 tokens out

```
{  
  s := "+-----+\n| Dave |\n+-----+\n";  
}
```

Final Program

```
method p_1_5_box() returns (s: string)  
  ensures s == "+-----+\n| Dave |\n+-----+\n"  
{  
  s := "+-----+\n| Dave |\n+-----+\n";  
}
```

Total Token Usage

Input tokens: 294

Output tokens: 97

Reasoning tokens: 57

Sum of ‘total tokens’: 391

Experiment Timings

Overall Experiment started at 1780483374001, ended at 1780483380412, lasting 6411ms (6.41 seconds)

Iteration #1 started at 1780483374002, ended at 1780483380412, lasting 6410ms (6.41 seconds)

Experiment 'p_1_6_large_letters' Results

June 3, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_1_6_large_letters

Natural language statement: Write a method that returns the name Harry in large letters.

Method signature: p_1_6_large_letters() returns (s: string)

Ensures

- `s == "* * * ***** * *\n* * * * * * * *\n***** * * ***** * *\n* * * * * * * *"`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method that returns the name Harry in large letters.

The signature should be:

p_1_6_large_letters() returns (s: string)

The method should respect the following contract:

```
ensures s == "* * * ***** ***** * *\n* * * * * * * * *\n***** * * ***** ***** * *\n* *  
↪ ***** * * * * *\n* * * * * * * * *"
```

Produce and show only the Dafny body of this method, including the curly braces that
↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor
↪ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DmdLpMYai2RLBgufsHITI8nj9Uunk

Model: gpt-5.5-2026-04-23

Created at: 1780483381

Finish reason: stop

Usage: 406 tokens in, and 222 tokens out

```
{  
  s := "* * * ***** ***** * *\n* * * * * * * * *\n***** * * ***** ***** * *\n* *  
↪ ***** * * * * *\n* * * * * * * * *";  
}
```

Final Program

```
method p_1_6_large_letters() returns (s: string)  
  ensures s == "* * * ***** ***** * *\n* * * * * * * * *\n***** * * ***** ***** * *\n* *  
↪ ***** * * * * *\n* * * * * * * * *"  
{  
  s := "* * * ***** ***** * *\n* * * * * * * * *\n***** * * ***** ***** * *\n* *  
↪ ***** * * * * *\n* * * * * * * * *";  
}
```

Total Token Usage

Input tokens: 406

Output tokens: 222

Reasoning tokens: 111

Sum of 'total tokens': 628

Experiment Timings

Overall Experiment started at 1780483381298, ended at 1780483389980, lasting 8682ms (8.68 seconds)

Iteration #1 started at 1780483381299, ended at 1780483389980, lasting 8681ms (8.68 seconds)

Experiment 'p_1_7_face' Results

June 3, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_1_7_face

Natural language statement: Write a method that returns a face in ASCII art.

Method signature: p_1_7_face() returns (s: string)

Ensures

- `s == " ////\n +\"\"\"\"\"\"+\\n(| o o |)\n | ^ |\n | ' - ' |\n +-----+\n"`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method that returns a face in ASCII art.

The signature should be:

p_1_7_face() returns (s: string)

The method should respect the following contract:

ensures s == " /////\n +\"\"\"\"\"\"+\\n(| o o |)\n | ^ |\n | ‘’- |\n +-----+\n"

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DmdLyVH235EPyXp9fhdfAqhRnrBsi

Model: gpt-5.5-2026-04-23

Created at: 1780483390

Finish reason: stop

Usage: 304 tokens in, and 177 tokens out

```
{  
  s := " /////\n +\"\"\"\"\"\"+\\n(| o o |)\n | ^ |\n | ‘’- |\n +-----+\n";  
}
```

Final Program

```
method p_1_7_face() returns (s: string)  
  ensures s == " /////\n +\"\"\"\"\"\"+\\n(| o o |)\n | ^ |\n | ‘’- |\n +-----+\n"  
{  
  s := " /////\n +\"\"\"\"\"\"+\\n(| o o |)\n | ^ |\n | ‘’- |\n +-----+\n";  
}
```

Total Token Usage

Input tokens: 304

Output tokens: 177

Reasoning tokens: 106

Sum of ‘total tokens’: 481

Experiment Timings

Overall Experiment started at 1780483390792, ended at 1780483398029, lasting 7237ms (7.24 seconds)

Iteration #1 started at 1780483390793, ended at 1780483398029, lasting 7236ms (7.24 seconds)

Experiment ‘p_1_8_painting’ Results

June 3, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 2

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 2

Problem Specification

Problem name: p_1_8_painting

Natural language statement: Write a method that returns an imitation of a Piet Mondrian painting. (Search the Internet if you are not familiar with his paintings.) Use character sequences such as @@@ or ::: to indicate different colors, and use - and | to form lines.

Method signature: p_1_8_painting() returns (s: string)

Ensures

- `s == mystring(" !!!@@@@@@@@@@@@@@@@@@@@\n", 5)`

Functional Code Given

```
function mystring(s: string, n: nat): string
{
  if n == 0 then "" else s + mystring(s, n - 1)
}
```

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method that returns an imitation of a Piet Mondrian painting. (Search the
↪ Internet if you are not familiar with his paintings.) Use character sequences such
↪ as @@@ or ::: to indicate different colors, and use - and | to form lines.

The signature should be:

p_1_8_painting() returns (s: string)

The method should respect the following contract:

ensures s == mystring(" !!!@@@@@@@@@@@@@@@@@@@@@@@@\n", 5)

The contract uses the following dafny code:

```
function mystring(s: string, n: nat): string
{
  if n == 0 then "" else s + mystring(s, n - 1)
}
```

These function(s) must not be used in your implementation of the method.

Produce and show only the Dafny body of this method, including the curly braces that
↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor
↪ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DmdM7vPg45Gj0l594erVXTnxziAvK

Model: gpt-5.5-2026-04-23

Created at: 1780483399

Finish reason: stop

Usage: 460 tokens in, and 574 tokens out

```
{
  s := " !!!@@@@@@@@@@@@@@@@@@@@@@@@\n !!!@@@@@@@@@@@@@@@@@@@@@@@@\n !!!@@@@@@@@@@@@@@@@@@@@@@@@\n
↪ !!!@@@@@@@@@@@@@@@@@@@@@@@@\n !!!@@@@@@@@@@@@@@@@@@@@@@@@\n";
}
```

Program → GenAI

When using dafny verify, the below error is emitted and verify fails:

```
problem_data/p_1_8_painting/program.dfy(8,0): Error: a postcondition could not be proved
  ↪ on this return path
```

```
|
8 | {
| ^
```

```
problem_data/p_1_8_painting/program.dfy(7,11): Related location: this is the
  ↪ postcondition that could not be proved
```

```
|
7 | ensures s == mystring(" !!!@@@@@@@@@@@@@@@@@@@@@\\n", 5)
| ^^
```

Dafny program verifier finished with 2 verified, 1 error

Correct the error in code by altering only the method body. Produce and show only the
↪ Dafny body, including the curly braces that surround it. Do not show the signature
↪ nor contract. You must not use 'assume' nor 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DmdMPMPmV4zF9UaZxDkRi2GrPrupe

Model: gpt-5.5-2026-04-23

Created at: 1780483417

Finish reason: stop

Usage: 683 tokens in, and 2872 tokens out

```
{
  var line := " !!!@@@@@@@@@@@@@@@@@@@@@\\n";
  s := line + (line + (line + (line + (line + ""))));

  assert mystring(line, 0) == "";
  assert mystring(line, 1) == line + mystring(line, 0);
  assert mystring(line, 1) == line + "";
  assert mystring(line, 2) == line + mystring(line, 1);
  assert mystring(line, 2) == line + (line + "");
  assert mystring(line, 3) == line + mystring(line, 2);
  assert mystring(line, 3) == line + (line + (line + ""));
  assert mystring(line, 4) == line + mystring(line, 3);
  assert mystring(line, 4) == line + (line + (line + (line + "")));
  assert mystring(line, 5) == line + mystring(line, 4);
  assert mystring(line, 5) == line + (line + (line + (line + (line + ""))));
}
```

Final Program

```
function mystring(s: string, n: nat): string
{
  if n == 0 then "" else s + mystring(s, n - 1)
```

```

}

method p_1_8_painting() returns (s: string)
  ensures s == mystring(" !!!@@@@@@@@@@@@@@@@@@@@\n", 5)
{
  var line := " !!!@@@@@@@@@@@@@@@@@@@@\n";
  s := line + (line + (line + (line + (line + ""))));

  assert mystring(line, 0) == "";
  assert mystring(line, 1) == line + mystring(line, 0);
  assert mystring(line, 1) == line + "";
  assert mystring(line, 2) == line + mystring(line, 1);
  assert mystring(line, 2) == line + (line + "");
  assert mystring(line, 3) == line + mystring(line, 2);
  assert mystring(line, 3) == line + (line + (line + ""));
  assert mystring(line, 4) == line + mystring(line, 3);
  assert mystring(line, 4) == line + (line + (line + (line + "")));
  assert mystring(line, 5) == line + mystring(line, 4);
  assert mystring(line, 5) == line + (line + (line + (line + (line + ""))));
}

```

Total Token Usage

Input tokens: 1143

Output tokens: 3446

Reasoning tokens: 3072

Sum of ‘total tokens’: 4589

Experiment Timings

Overall Experiment started at 1780483398699, ended at 1780483484370, lasting 85671ms (85.67 seconds)

Iteration #1 started at 1780483398699, ended at 1780483417349, lasting 18650ms (18.65 seconds)

Iteration #2 started at 1780483417349, ended at 1780483484370, lasting 67021ms (67.02 seconds)

Experiment 'p_1_9_house' Results

June 3, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_1_9_house

Natural language statement: Write a method that returns a house in ASCII art.

Method signature: p_1_9_house() returns (s: string)

Ensures

- `s == "\n + \n + +\n+-----+\n| .-. |\n| | | |\n+---+++\n"`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method that returns a house in ASCII art.

The signature should be:

`p_1_9_house()` returns (`s: string`)

The method should respect the following contract:

`ensures s == " +\n + +\n + +\n+-----+\n| .-. |\n| | | |\n+---+++\n"`

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DmdNVWvZxswukf6POjjij3qQcX7tM

Model: gpt-5.5-2026-04-23

Created at: 1780483485

Finish reason: stop

Usage: 300 tokens in, and 57 tokens out

```
{  
  s := " +\n + +\n + +\n+-----+\n| .-. |\n| | | |\n+---+++\n";  
}
```

Final Program

```
method p_1_9_house() returns (s: string)  
  ensures s == " +\n + +\n + +\n+-----+\n| .-. |\n| | | |\n+---+++\n"  
{  
  s := " +\n + +\n + +\n+-----+\n| .-. |\n| | | |\n+---+++\n";  
}
```

Total Token Usage

Input tokens: 300

Output tokens: 57

Reasoning tokens: 0

Sum of ‘total tokens’: 357

Experiment Timings

Overall Experiment started at 1780483485111, ended at 1780483490623, lasting 5512ms (5.51 seconds)

Iteration #1 started at 1780483485111, ended at 1780483490623, lasting 5512ms (5.51 seconds)

